

Original Investigation

Systematic Growth Monitoring for the Early Detection of Celiac Disease in Children

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IMPORTANCE Growth-monitoring programs in children aim to achieve the early detection of disorders that affect growth. Celiac disease (CD) is underdiagnosed in the pediatric population in which the presenting features often include faltering linear growth, short stature, or poor weight gain.

OBJECTIVES To develop new evidence-based cutoffs for screening for growth disorders and to evaluate the performance of these cutoffs among children with CD measured regularly in a nationwide growth screening program.

DESIGN, SETTING, AND PARTICIPANTS A longitudinal retrospective study that included longitudinal growth data of healthy children (the reference population) from primary health care and children with CD (the cases) from primary health care and 3 university hospital outpatient clinics in Finland (Kuopio University Hospital, Tampere University Hospital, and Helsinki University Hospital) from January 1, 1994, to April 9, 2009. Children of the reference population were between 0 and 20 years of age and children with CD were between 1 and 16 years of age. In the reference population of 51 332 healthy children, 5 age-specific and sex-specific growth-screening parameters (height standard deviation score and body mass index standard deviation score distance from the population mean, distance from target height, change in height standard deviation score, and change in body mass index standard deviation score) were developed. Performance of these parameters and their combination was evaluated in 177 children with CD by analyzing longitudinal growth data from birth until diagnosis of CD.

MAIN OUTCOME AND MEASURE The screening accuracy for detecting abnormal growth in children with CD, assessed using receiver operating characteristics analysis expressed as the area under the curve.

RESULTS Celiac disease was detected with good accuracy (area under the curve [95% CI] = 0.88 [0.84–0.93] for girls and 0.84 [0.77–0.91] for boys) when screening was performed using the combination of all 5 growth-screening parameters. When the specificity of the screening was set at 90%, growth was already abnormal in 57% of the girls with CD and 48% of the boys with CD 2 years prior to diagnosis.

CONCLUSIONS AND RELEVANCE Prior to diagnosis, growth faltered in most children with CD. These children could have been detected several years earlier by a well-established growth-monitoring program. Acceptable screening accuracy can be achieved for CD via the use of several growth-monitoring parameters in combination, preferably using computerized screening algorithms that are integrated into an electronic health record system.

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Growth-monitoring programs have been a fundamental part of preventive child health care for more than a century.¹ Ideally, screens for abnormal growth aim to detect treatable illnesses before any other symptoms of the disease appear. Early diagnosis helps to minimize the impact of the underlying condition and optimize final adult height.^{2,3} During growth screening, height and weight values should be compared with the pre-established cutoff values that define normal growth.⁴ Despite the long tradition of growth monitoring, evidence-based cutoffs for clinical pediatric practice are insufficiently explored and studies that aim to define such norms are scarce.²⁻⁵

Celiac disease (CD) is an immune-mediated systemic disorder elicited by gluten and related prolamines and characterized by a variety of nonspecific signs and symptoms including faltering linear growth, short stature, and poor weight gain in children.⁶ Celiac disease remains underdiagnosed during childhood; based on serological screening studies in both symptomatic CD and subclinical CD, this disease has a prevalence of more than 1% to 2% in Western populations^{7,8} but only 10% to 30% of these patients are diagnosed.⁶⁻¹² Universal serological screening for CD in the child population is an option for improving the detection rate of CD but the benefits and cost-effectiveness of such screening remain controversial.¹²⁻¹⁵ Thus, universal screening is currently not recommended by the current consensus statements for the diagnosis of CD in Europe and the United States.^{6,10,12} These guidelines recommend serological tests for tissue transglutaminase antibodies to rule out CD in children with symptoms suggestive of CD, such as growth failure.^{6,9,11} However, criteria for defining growth failure in children with CD are not included in any of the guidelines mentioned earlier.

Auxological screening of CD has not been extensively studied.^{3,16} The aim of the current study was to develop evidenced-based cutoffs for screening for growth disorders and test the performance of these cutoffs in the screening of children with CD.

Methods

This study was approved by the ethics committee of the Pohjois-Savo Health Care District. No contact was made with study participants and consent was not provided owing to the retrospective nature of the study; all data were analyzed anonymously.

Growth Reference Population

Mixed cross-sectional and longitudinal growth data from 51 332 healthy Finnish children (children with known medical diagnoses or medications possibly interfering with growth were removed; n = 2351) aged 0 to 20 years (25 059 girls and 26 273 boys), including 191 467 measurements, were collected for the construction of a contemporary nationwide Finnish growth reference.¹⁷ These data comprised the reference population data in this study. For the growth reference, healthy children were measured during routine visits at well-baby clinics or school health care as a part of a nationwide free growth-

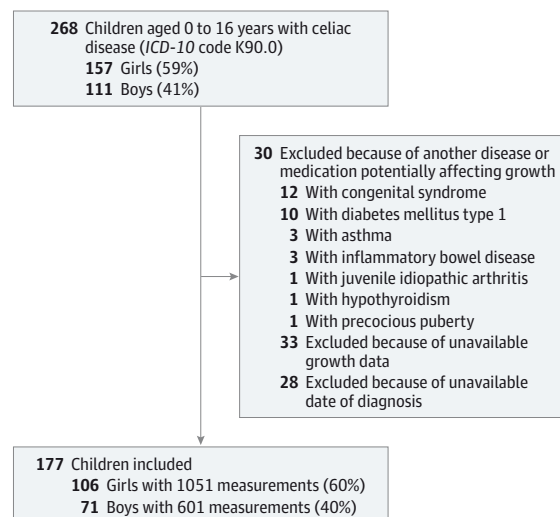
monitoring program that has been used for more than 20 years in Finland. The measurements were captured using an electronic patient management system. The reference population is described in detail elsewhere.¹⁷

Children With CD

Children aged 0 to 16 years with CD were identified in the patient registries of 3 university hospitals in Finland (Helsinki, Kuopio, and Tampere) according to the *International Statistical Classification of Diseases, Tenth Revision* code for CD (K90.0). Clinical data were collected retrospectively from patient files at the hospitals. Similar to almost all Finnish children, these children with CD participated in the nationwide free-of-charge growth-monitoring program and had been measured during every routine visit at well-baby clinics or school health care. These auxological data were obtained, registered, and combined with the growth data obtained using similar equipment and measurement techniques at the hospital visits. However, most of the measures were performed in primary care. Growth data were collected until the time of diagnosis. Height and weight measurements were transformed into z scores according to the new Finnish growth reference.¹⁷ Potentially false measurements, typing errors, missing values, and duplicated recordings were evaluated using scatterplots and were then either corrected or excluded. In all cases, the diagnosis of CD was based on the histological evaluation of duodenal biopsies by a pathologist. The biopsy specimens were classified as having either moderate, subtotal, or total villous atrophy. In 5 cases, the degree of villous atrophy was unavailable. The age at diagnosis, preceding symptoms, laboratory results (ie, hemoglobin, total IgA, IgA tissue transglutaminase, and endomysial antibody levels), and histological data were registered.

The initial study cohort consisted of 268 children (59% girls; Figure 1). After excluding children with another disease or

Figure 1. Flowchart of the Study Cohort



ICD-10 indicates *International Statistical Classification of Disease, Tenth Revision*.

Table 1. Clinical Characteristics of 177 Children With Celiac Disease at the Time of Diagnosis

Variable	Girls	Boys
Total, No. (%)	106 (60)	71 (40)
Height and weight		
Measurements, No.	1051	601
Measurements per child, mean (SD)	9.9 (6.0)	8.5 (4.8)
Height for age SDS, mean (SD) ^a	-0.45 (1.1)	-0.58 (1.2)
Height distance from population, mean (SD), cm ^a	-1.9 (4.9)	-2.6 (6.8)
BMI for age SDS, mean (SD) ^a	-0.25 (1.2)	-0.44 (1.1)
Target height SDS, mean (SD) ^b	-0.05 (0.6)	-0.06 (0.8)
Height distance from target height SDS, mean (SD)	-0.57 (1.2)	-0.67 (1.1)
Age at diagnosis, median (range), y	6.2 (0.9-15.9)	7.1 (0.8-16.1)
Presenting symptoms, No. (%)		
Diarrhea and abdominal pain	20 (19)	25 (35)
Constipation and abdominal pain	10 (9)	2 (3)
Only abdominal pain	28 (26)	11 (16)
Nonspecific symptoms or positive family history	30 (28)	18 (25)
Not reported	18 (17)	15 (21)
Histological findings in duodenal biopsy		
Villous atrophy, No. (%)		
Moderate	39 (37)	22 (31)
Subtotal	47 (44)	26 (37)
Total	15 (14)	23 (32)
Unavailable, No. (%)	5 (5)	0 (0)
Laboratory findings, median (range)		
IgA-tTG, U/mL	166 (17-498)	167 (15-1843)
IgA-EMA, titer	250 (5000)	250 (10 240)
Hb, g/dL	12.2 (8.7-13.9)	12.2 (8.0-16.1)

Abbreviations: BMI, body mass index (calculated as weight in kilograms divided by height in meters squared); IgA-EMA, immunoglobulin A endomysium antibody; IgA-tTG, immunoglobulin A tissue transglutaminase antibody; Hb, hemoglobin; SDS, standard deviation score.

SI conversion factors: to convert IgA to milligrams per liter, multiply by 10 and to convert hemoglobin to grams per liter, multiply by 10.0.

^a In comparison with the population-specific Finnish growth reference.¹⁷

^b Target height formula: $0.791 \times \text{mean parental height SDS} - 0.147$ for girls and $0.886 \times \text{mean parental height SDS} - 0.071$ for boys.⁵

medication that could affect growth, children with unavailable growth data (patients with CD who were only biopsied at the hospital without auxological recordings) and children with an unavailable date of CD diagnosis, the final study cohort consisted of 177 children, including 106 girls (60%) and 71 boys with 1051 height measurements and 601 weight measurements prior to diagnosis, respectively. On average, children with CD were measured 9 times whereas children in the reference population were measured 4 times ($P < .001$). Auxological data were available for 68 girls and 39 boys with CD as many as 5 years prior the diagnosis. The median age at diagnosis was 6.2 years (range, 0.9-15.9 years) in girls and 7.1 years (range, 0.8-16.1 years) in boys ($P = .14$; Table 1). Parental heights were available for all children with CD.

Growth-Monitoring Parameters and Statistical Analysis

The 5 new growth-screening parameters tested in screens for CD were the height for age standard deviation score (HSDS), body mass index (BMI; calculated as weight in kilograms divided by height in meters squared) for age SDS (BMI SDS), HSDS distance from target height (TH) (TH formula = $0.791 \times \text{mean parental height SDS} - 0.147$ for girls and $0.886 \times \text{mean parental height SDS} - 0.071$ for boys⁵), change in height SDS (Δ HSDS), and change in BMI SDS (Δ BMI SDS) across time with free age intervals. These parameters were used during screening individually or as a combination of all 4 parameters.

The calculations of the age-specific and sex-specific normal values for Δ HSDS and Δ BMI SDS were based on the longitudinal measurements of the reference population. The statistical methods used are described in detail in the eAppendix and eTable in the Supplement. A mixed linear model for repeated measurements was used to test the difference in the HSDS of children with CD compared with the reference population mean height.

The diagnostic accuracy of the growth-screening parameters was depicted using the area under the curve (AUC) values from a receiver operating characteristic analysis that was performed compared with the reference population. The receiver operating characteristic curve analyses were based on the individual probability for abnormal growth, which was defined for the HSDS, BMI SDS, TH SDS, Δ HSDS, Δ BMI SDS, or a combination of these parameters and was calculated for each child in the CD cohort and the reference population using logistic regression. The results of the receiver operating characteristic analyses were classified based on AUC values as fail (0.50-0.59), poor (0.60-0.69), moderate (0.70-0.79), good (0.80-0.89), and excellent (0.90-1.00). The diagnostic performance of the combined parameters was also evaluated based on the time elapsed from the first abnormal screening result until the clinical diagnosis of CD.

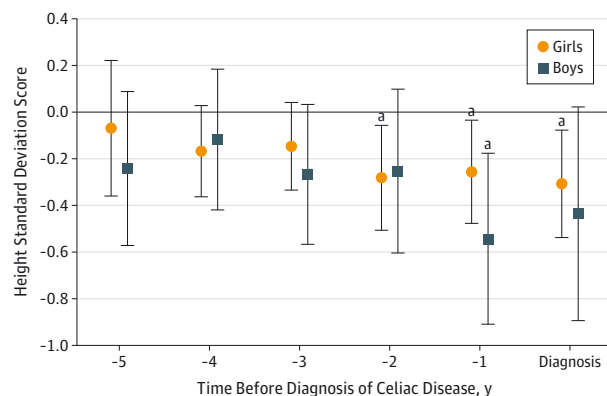
The data were analyzed using R statistical software version 3.0.1 (The R Project for Statistical Computing) and SPSS software, version 19 (IBM Corporation). *P* values less than .05 were considered statistically significant.

Results

At the time of diagnosis, the most common symptom preceding the CD diagnosis was recurrent abdominal pain, which occurred in 27% of the children. The histopathological findings revealed a more severe grade of atrophy in boys, with total villous atrophy in 23 boys (32%) in comparison with 15 girls (14%) ($P < .01$). Most children with CD had an increased level of IgA antibodies against tissue transglutaminase (median, 166 mg/dL in girls and 167 mg/dL in boys) and endomysial antibody (titer of 767 in girls and titer of 992 in boys). The IgA-tissue transglutaminase and endomysial antibodies were in the normal range in 9 patients and 2 of these children also exhibited total IgA deficiency. At the time of diagnosis, hemoglobin was less than the age-specific reference range in 31% of children with CD.

In the CD cohort at the time of diagnosis, the mean (SD) HSDS and BMI SDS were -0.45 (1.08) and -0.25 (1.23), respec-

Figure 2. Mean Height Standard Deviation Score (95% CI) in Girls and Boys With Celiac Disease 0 to 5 Years Prior to Diagnosis in Comparison With the Population-Based Reference¹⁷



The population-based reference indicated as the zero line.

^aStatistically significant difference from the reference population ($P < .05$).

tively, in girls and -0.58 (1.17) and -0.44 (1.08) in boys, respectively (Table 1). Overall, girls with CD were shorter than the reference population 2 years prior to the CD diagnosis and boys were shorter than the reference population 1 year prior to the CD diagnosis ($P < .05$; Figure 2). However, at the time of diagnosis, the boys were not shorter than the reference population, on average ($P = .06$). In addition, the observed differences in BMI SDS between patients with CD and the reference population were not statistically significant for girls or boys (data not shown).

Screening for abnormal growth in patients with CD using any single screening parameter resulted in poor to moderate accuracy (Figure 3). The screening accuracy (AUC) for the Δ HSDS parameter was 0.71 (95% CI, 0.67-0.77) in girls and the screening accuracy for the Δ BMI SDS parameter was 0.72 (95% CI, 0.64-0.80) in boys. The AUCs for HSDS, BMI SDS, and TH SDS varied between 0.60 and 0.68 in girls and 0.64 and 0.67 in boys. The 5 screening parameters in combination (ie, ≥ 1 abnormal screening parameters for a child) performed better than any of the parameters alone. The AUC for the combination of the parameters was 0.88 (95% CI, 0.84-0.93) in girls and 0.84 (95% CI, 0.77-0.91) in boys.

The accuracies of each of the growth screening parameters separately and in combination for detecting abnormal growth as many as 5 years preceding the diagnosis are presented in Table 2. Altogether, sensitivity of the screening when growth parameters were used in combination was 69% for girls and 61% for boys at a 90% specificity level (ie, 10% of children with an abnormal screening result would not have CD). The sensitivity at a specificity level of 95% was 49% for girls and 51% for boys and the sensitivity at a specificity level of 99% was 26% for both sexes.

In the analysis of the longitudinal growth data from the 5 years preceding the CD diagnosis, the cumulative prevalence of an abnormal screening result using the 5 screening parameters in combination increased as the CD diagnosis moved closer (Figure 4). If the specificity level of the growth screen-

ing was set at 90%, the cumulative percentage of children with abnormal growth prior to or at CD diagnosis was 82% for girls and 70% for boys. The corresponding percentages at specificity levels of 95% and 99% were 66% and 39% for girls and 65% and 39% for boys, respectively. Abnormal growth was already observed 2 years prior to the CD diagnosis in 57% of girls and 48% of boys at a specificity level of 90%. The median (range) delay between the first abnormal result using any of the growth screening rules and the CD diagnosis was 3.2 years (0-4.9 years) in girls and 2.7 years (0-4.9 years) in boys at a specificity level of 90% (data not shown). If the specificity of the screening was set at 95% or 99%, the median (range) delay was 3.0 years (0-5.0 years) or 2.6 years (0-5.0 years) in girls, respectively, and 2.3 years (0-5.0 years) or 2.2 years (0-5.0 years) in boys, respectively.

Discussion

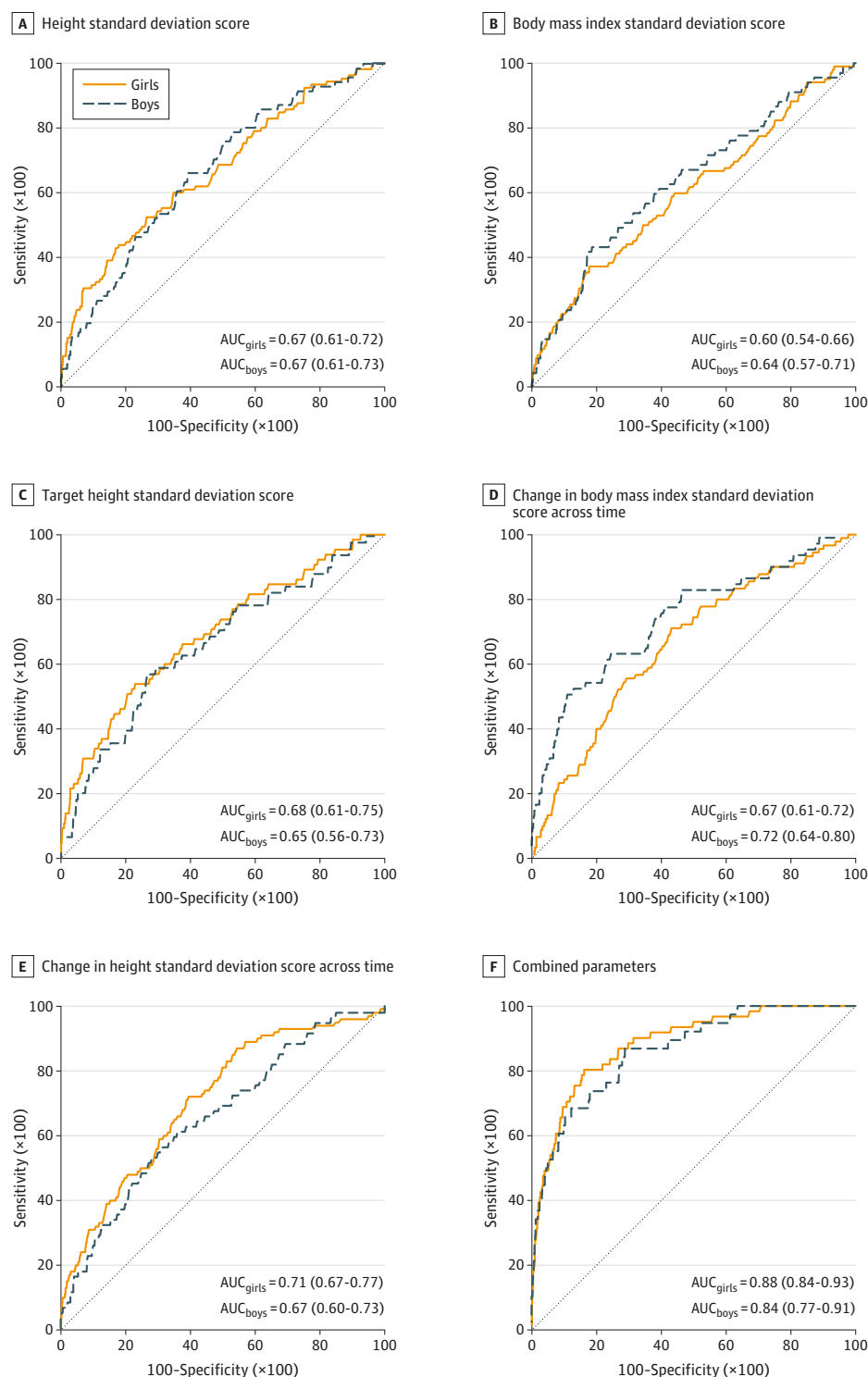
In this study, we showed that most children diagnosed as having CD would have been detected by auxological screening, which is a simple and noninvasive method for improving the early diagnosis of CD in children. Most children with CD developed a growth failure prior to the diagnosis and, on average, children with CD were shorter than the healthy reference population. The best growth screening accuracy can be obtained if childhood growth is monitored longitudinally rather than via one-off measurements and if the height and weight measurements are systematically evaluated using established screening parameters and contemporary cutoff values.

The major strengths of this study were the use of a population-based contemporary growth reference and defined cutoff values for specific parameters, including the change in height SDS and BMI SDS across time, which enabled the assessment of subtle changes in the growth rate of children with CD. Our finding in small changes in the growth rate in children with CD is convergent with Jansen et al.¹⁸ Few previous studies have assessed the use of growth rate for screening.^{3,5}

The limitations of this study were its retrospective nature and the collection of data from patient files. The best approach would be a prospective population-based screening study in which growth-screening parameters were compared with serological screening tests for CD. In addition, patient files did not include systematic notes on intestinal and extraintestinal symptoms prior to the diagnosis of CD. However, studies from England⁷ and Sweden⁸ indicated that active case finding based on symptoms of CD rarely facilitated early diagnosis. Those authors performed serological screening for CD in an unselected population of children of 7.5 and 12 years of age who were also screened for related symptoms via questionnaires. In most children with CD detected in these studies, no CD-associated symptoms or conditions existed.

We believe that the retrospective growth data were not biased. The growth data were retrieved from electronic health records and all of the patients with CD were measured regularly as part of the nationwide growth-monitoring program that has been operative since the 1970s.¹⁹⁻²¹ The quality

Figure 3. Diagnostic Accuracy of 5 Growth-Monitoring Parameters and Their Combination in 106 Girls and 71 Boys With Celiac Disease



of the growth data gathered at routine visits has also been assessed; false measurements, typing errors, missing values, or duplicate recordings are scarce.²² However, children with CD were measured more frequently than children in the reference population.

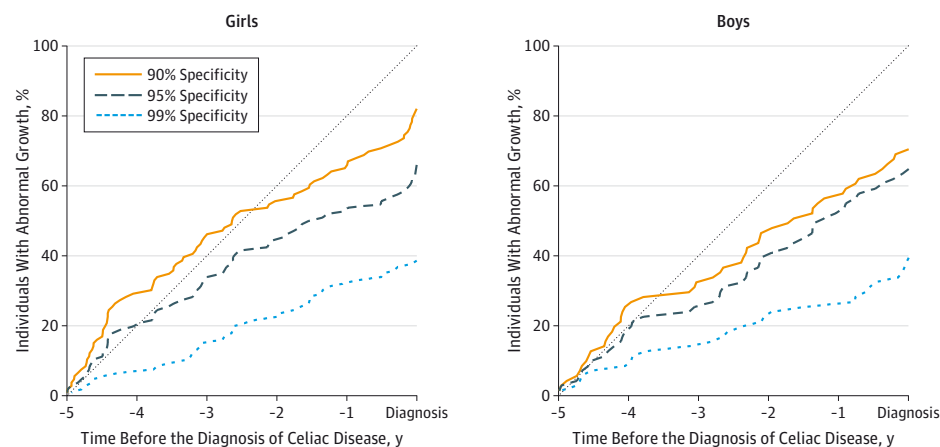
The clinical pattern of childhood CD has changed from the classic triad (ie, failure to thrive, diarrhea, and abdominal distention) to a variety of nonspecific signs and symptoms,^{6,23-26} as observed in the present cohort. Abnormal growth is the most common extraintestinal symptom of CD.²⁷ The probability of

Table 2. Sensitivity of Individual and Combined Growth-Monitoring Parameters for Detecting Abnormal Growth in Children With Celiac Disease

Growth-Monitoring Parameter	Mean (95% CI)					
	90 % Specificity		95 % Specificity		99 % Specificity	
	Girls	Boys	Girls	Boys	Girls	Boys
Height standard deviation score	31 (23-41)	25 (16-37)	24 (16-33)	15 (8-26)	10 (5-17)	6 (2-14)
Body mass index ^a standard deviation score	23 (15-32)	22 (13-35)	14 (8-23)	15 (8-26)	7 (3-14)	5 (1-13)
Target height standard deviation score	31 (20-43)	26 (15-41)	23 (14-35)	16 (7-30)	11 (4-21)	2 (0-11)
Change in height standard deviation score across time	31 (22-41)	27 (17-40)	20 (13-29)	17 (9-29)	10 (5-18)	8 (3-18)
Change in body mass index standard deviation score across time	24 (16-35)	41 (28-56)	13 (7-22)	24 (13-38)	7 (2-14)	10 (3-22)
Combined	69 (56-80)	61 (45-77)	49 (36-62)	51 (35-68)	26 (16-39)	26 (13-42)

^a Calculated as weight in kilograms divided by height in meters squared.

Figure 4. Cumulative Percentage of Children With Abnormal Growth (by any of the 5 or Combined Growth-Monitoring Parameters) 0 to 5 Years Prior to the Diagnosis of Celiac Disease



CD is estimated to range from 19% to 59% in children with a short stature of nonendocrinological cause.²⁸⁻³⁰ Nevertheless, severe growth failure is currently uncommon in children with CD, particularly in developed countries.²⁹⁻³¹ In contrast to our observations, growth failure in contemporary children with CD was not observed in the study performed by Savilahti et al.³² This discrepancy likely occurred owing to the fact that growth failure is relatively subtle in nature; because those authors used outdated growth references to evaluate growth, secular changes in height caused a systematic error in their data. Our findings of impaired growth in patients with CD, which was observed 2 years prior to diagnosis in girls and 1 year prior to diagnosis in boys, are consistent with the study reported by Korponay-Szabo et al.¹³ Those authors performed universal serological screening for CD among 6-year-old children in 1 county ($n = 2360$) and diagnosed 32 new CD cases. These 32 children were, on average, shorter and lighter than their healthy peers.¹³ In our study, boys were not shorter than girls at the time of diagnosis, which is best explained by the relatively small number of children with CD.

Consistent with our results, van Dommelen et al¹⁶ recently showed in a Dutch cohort that the change in BMI SDS was the most accurate parameter for detecting CD. However, the Dutch cohort comprised primarily infants and small children (0-2.5 years of age), while the children in our study ranged

in age from infancy to late puberty; therefore, the 2 studies are not directly comparable. We found that the best screening accuracy for CD was accomplished by combining various growth-screening parameters. Nevertheless, the screening accuracy for CD still remained inferior to the screening accuracy achieved for other disorders that affected growth, such as Turner syndrome.^{4,5} The diagnostic accuracy of any screening program is always a trade-off between sensitivity and specificity. Growth-monitoring cutoffs are intended to detect as many abnormally growing children as possible (ie, high sensitivity) without producing too many unnecessary referrals or further investigations (ie, high specificity). In that respect, a relatively low specificity can be considered acceptable for CD screening because the detected children are investigated further using simple and inexpensive serological testing. However, detecting as many patients with CD as possible by growth monitoring (ie, higher sensitivity) always leads to examinations of children without CD as well (lower specificity).

Recent data demonstrated that CD remains severely underdiagnosed in Western populations⁶⁻¹¹; therefore, new tools are needed to improve diagnosis. Universal population-based screening for CD using serology is not currently recommended^{6,10,12} but growth monitoring is widely accepted and performed in nearly every developed country. Improvements in systematic growth screening using pre-

established rules would facilitate more efficacious screening for disorders such as CD. This principle was demonstrated in our study by the existence of a significant diagnostic delay between the first abnormal height or weight measurement and the diagnosis of CD. Screening accuracy might be even more effective in populations with a high relative risk for CD. In addition, the use of a population-based growth reference rather than a universal reference facilitates the identification of subtle forms of growth failure, as shown previously in the context of Turner syndrome.³³ We recently demonstrated that the implementation of a growth-monitoring program into electronic health records and the computerization of the screening process is a feasible and effective method for improving the early diagnosis of growth disorders.³⁴ Consequently, we recommend systematic growth screening as a primary method for the early detection of chronic disorders that affect growth, such as CD. A prospective population-

based study is warranted to evaluate the benefits and costs of such a screening program.

Conclusions

Growth failure remains an early and common feature in patients with CD and an up-to-date growth reference and well-established growth-monitoring program could facilitate the early diagnosis of CD. In addition, population-based screening for CD can be performed with good accuracy when several screening parameters for abnormal growth are used simultaneously in combination with the use of longitudinal growth data. Owing to the complex nature of evidence-based growth screening, this process should ideally be performed using computerized screening algorithms integrated into electronic health record systems.

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Acquisition, analysis, or interpretation of data: Saari, Harju, Mäkitie, Saha, Dunkel, Sankilampi.

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